Finite calculus: summation by parts, and the Euclidean algorithm

Notes from Carlos' lessons

Source image (undated, filed in `misc/`): finite calculus and Euclid's GCD. The file's timestamp is 2026-04-16; the content matches the first chapters of *Concrete Mathematics* (difference calculus) and the Euclidean algorithm.

1 Difference operator and product rule

Let $\Delta f(x) = f(x+1) - f(x)$ and let E be the shift, $Ef(x) = f(x+1)$.

Theorem 1 (Product rule)**.** $\Delta(fg) = (\Delta f)\, g + E(f)\, (\Delta g)$.

Proof.

$$f(x{+}1)g(x{+}1) - f(x)g(x) = \big(f(x{+}1) - f(x)\big)g(x) + f(x{+}1)\big(g(x{+}1) - g(x)\big).$$

□

Summing (telescoping) both sides over a range gives **summation by parts**:

$$\sum (\Delta f)\, g\ \delta x = fg - \sum E(f)\, (\Delta g)\ \delta x$$

(the board writes the indefinite sums as $\sum \ldots \delta x$ in the boxed line).

2 Exponentials, and an example

$\Delta(b^x) = b^{x+1} - b^x = (b-1)\, b^x$. In particular $\Delta(2^x) = 2^x$, so 2^x plays the role of e^x (it is its own difference).

Example: $\sum k\, 2^k\, \delta k$**.** Take $f = 2^k$ (so $\Delta f = 2^k$) and $g = k$ (so $\Delta g = 1$):

$$\sum k\, 2^k\, \delta k = k\, 2^k - \sum 2^{k+1} \cdot 1\, \delta k = k\, 2^k - 2 \cdot 2^k = (k-2)\, 2^k$$

(the board's chain: $k{\cdot}2^k - \sum 2^{k+1}$, then $k{\cdot}2^k - 2\sum 2^k$, then $k{\cdot}2^k - 2^{k+1}$). So $\sum_{k=0}^{n-1} k\, 2^k = (n-2)2^n + 2$. Check $n = 2$: $0 + 2 = 2$ and the formula gives $0 + 2 = 2$. Check $n = 3$: $0 + 2 + 8 = 10$ and the formula gives $(3-2) \cdot 8 + 2 = 10$.

3 The Euclidean algorithm (lower half of the board)

Let $D(a, b) = \gcd(a, b)$.

Claim 2 (Observation)**.** $D(a, b) = D(a - b,\, b) = D(a,\, b - a)$*, and more generally* $D(a, b) = D(a,\, b - qa)$ *for every integer* q*.*

Proof. The pairs (a, b) and $(a, b - qa)$ have the same set of common divisors: a number dividing a and b divides $b - qa$, and a number dividing a and $b - qa$ divides $(b - qa) + qa = b$. Same common divisors, so the same greatest one. □

In particular $D(a, b) = D(a,\, b \bmod a)$ with $0 \le b \bmod a \le a - 1$ (the board's "note").

Algorithm (the board's steps). (1) We have $D(a, b)$. (2) If $a > b$ go to $D(a \bmod b,\ b)$, else to $D(a,\ b \bmod a)$. Each step replaces the larger number by its remainder modulo the smaller one, so the pair strictly decreases; the process stops at $D(0, m) = m$. The last board line unrolls two steps: for $a < b$, $D(a, b) = D(a, b \bmod a) = D\big(a \bmod (b \bmod a),\ b \bmod a\big)$.

Notes

- A small yellow box (blank) and a teal tick in the middle of the image are leftover marks; nothing is written in them.